IPCON® PA6 CF

Carbon Fiber Reinforced Polyamide-6 (Nylon-6) Filament

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.05 mm
Net Filament Weight	1 kg ± 0.5%
Filament Length	360 m ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C) or PC (max heat resistance: 130 °C)
Spool Size	Outer Cardboard's diameter: 200 mm; Inner roller's diameter: 90 mm
Recommendation for body protection during use	This material contains carbon fiber, which may cause skin irritation. So it is recommended to wear gloves throughout the operation.

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.15 g/cm³
Shrinkage Rate*	Specially-defined	0.17% - 0.44%
Surface Hardness	Shore hardness	83D
Saturated Water Absorption Rate	25 °C, 55% RH, room air	2.08%
Melt Index	290 °C, 2.16 kg	14.7 g/10 min
Melting Temperature	DSC, 10 °C/min	226 °C
Vicar Softening Temperature	ISO 306, GB/T 1633	208 °C
Heat Deflection Temperature	ISO 75, 0.45 MPa	197 °C
Relative Thermal Index (RTI) Max Continuous Service Temperature	Actual working conditions simulation test	105 - 120 °C (dry state) (Related to load, size, shape, wall thickness, rate of water content, etc., and the value after annealing is higher than that before annealing in general)

Mechanical Properties*	Testing Methods	Data*
Tensile Strength XY	ISO 527, GB/T 1040	100 MPa
Tensile Strength Z	ISO 527, GB/T 1040	57 MPa
Young's Modulus XY	ISO 527, GB/T 1040	6173 MPa
Young's Modulus Z	ISO 527, GB/T 1040	2731 MPa
Breaking Elongation Rate XY	ISO 527, GB/T 1040	3.6%
Breaking Elongation Rate Z	ISO 527, GB/T 1040	1.9%
Flexural Strength XY	ISO 178, GB/T 9341	162 MPa

Flexural Strength Z	ISO 178, GB/T 9341	91 MPa
Flexural Modulus XY	ISO 178, GB/T 9341	5312 MPa
Flexural Modulus Z	ISO 178, GB/T 9341	2757 MPa
Impact Strength XY	ISO 179, GB/T 1043	50.5 kJ/m²
Impact Strength Z	ISO 179, GB/T 1043	14.7 kJ/m²

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu P1S printer (nozzle temperature: 290 °C, bed temperature: 100 °C, printing speed: 100 mm/s) and IPCON PA6 CF filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

1. Carbon fiber reinforced, ultra low warping, excellent layer strength, high toughness and impact resistance, high strength and stiffness, outstanding heat resistance; compatible with AMS and CFS.
2. Applications that require high heat resistance, light weight, high strength and high toughness, such as oven trays and racks, automobiles, drones, RC racing cars, power tools, etc.

• Recommended Printing Settings

Drying Settings Before Printing	100 - 120 °C, 8 - 14 h (Blast drying oven)
Printing Temperature and Humidity	≤ 70 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Compatible Support Material	Itself or support filament for PA
Compatible Printer Type	Enclosed-frame (recommended), open-frame,
AMS or CFS	Compatible
Compatible Nozzle Material	Hardened steel, diamond, tungsten carbide, etc
Compatible Nozzle Size	0.4 / 0.6 (recommended) / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plate
Plate Surface Preparation	Apply liquid glue or PVP solid glue stick if the first layer bonding is not firm enough
Nozzle Temperature	270 - 300 °C
Bed Temperature	80 - 100 °C
Fan Speed	0% - 40%
Max Print Speed	300 mm/s

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